

Q1: Select the most correct answer: (15 points)

1. What's the equivalent of $\$t2 = A[A[0]]$ in assembly where the base address of A is in $\$s0$:
- ☒ a. `lw $t0, 0($s0)`
`lw $t2, 0($t0)`
 - b. `lw $t0, 0($s0)`
`sll $t1, $t0, 2`
`add $t1, $s0, $t1`
`lw $t2, 0($t1)`
 - c. `lw $t0, 0($s0)`
`add $t1, $s0, $t0`
`lw $t2, 0($t1)`
 - d. `lw $t2, 0($s0)`
2. Which of the following instructions uses PC-relative addressing?
- a. `addi $t1, $t2, 10`
 - b. `lw $t0, 4($s0)`
 - c. `jr $ra`
 - d. `beq $t0, $t1, LABEL`
3. Assume a new instruction **move \$t0, \$t1** that copies the contents of $\$t1$ to $\$t0$. Which of the following standard MIPS instructions is **NOT** equivalent to this operation?
- a. `add $t0, $t1, $zero`
 - b. `addi $t0, $t1, 0`
 - c. `sub $t0, $t1, $zero`
 - ☒ d. `sub $t0, $zero, $t1`
4. Decode the machine instruction **0x01484820**:
- ☒ a. `add $t1, $t2, $t3`
 - b. `addi $t1, $t2, 20`
 - c. `and $t1, $t2, $t2`
 - d. `sub $t1, $t2, $t3`
5. Which of the following registers is used with one of J-type instructions?
- a. $\$t0$
 - c. $\$s0$
 - ☒ b. $\$ra$
 - d. $\$s3$
6. What's the final value stored in $\$t0$ after executing `lui $t0, 0x1234`, assuming $\$t0$ originally contains 0?
- a. 0x12340000
 - b. 0x00001234
 - c. 0xFFFF1234
 - ☒ d. 0x1234FFFF
7. If a `beq $t0, $t1, Label` at address 0x00400020 branches to 0x00400028, what is the value of Label?
- ☒ a. 1
 - b. 6
 - c. 4
 - d. 8

Student name:

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8. In which case can signed overflow occur in the operation $A - B$:

- a. A is positive and B is positive
- b. $A = 0$ and $B = 0$
- c. A is positive and B is negative
- d. A is negative and B is negative

9. Assume a 4-bits multiplicand and a 4-bits multiplier, what's the size of the ALU that could perform the first-version multiplication :

- a. 8-bits ALU
- b. 16-bits ALU
- c. 4-bits ALU
- d. 32-bits ALU

10. What is the IEEE 754 single-precision representation of the decimal value -1.5 ?

- a. 1011 1111 0110 0000 0000 0000 0000 0000
- b. 0100 0000 0100 0000 0000 0000 0000 0000
- c. 1000 0000 0110 0000 0000 0000 0000 0000
- d. 1011 1111 1100 0000 0000 0000 0000 0000

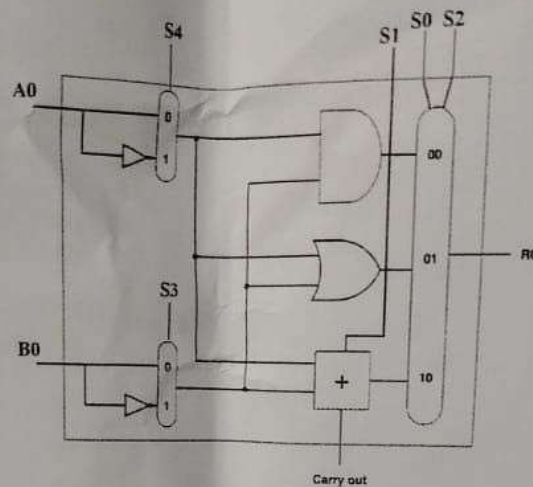
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10.5

Write your choice in the following Answers table. Other answers located outside the table will not be considered.

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
a	c	d	a	b	d	a	c	a	d

Q2: Based on Chapter 3 (ALU Design). Your goal is to understand the following design and answer the questions associated to it. Consider the distribution of the selectors (15 points)



1. [2.5 points] Write down input signals to perform subtraction ($A - B$):

S4	S3	S2	S1	S0
0	1	0	1	1

2. [3.5 points] Write down input signals to support a new instruction called "INC A" which increments A by 1 ($A = A + 1$):

A0	B0	S4	S3	S2	S1	S0
A	1	0	0	0	0	1

3. [3.5 points] Write down input signals to support a new instruction called "DEC A" which decrements A by 1 ($A = A - 1$):

A0	B0	S4	S3	S2	S1	S0
A	1	0	1	0	1	1

4. [3.5 points] Write down input signals to support a new instruction called "INV A" which inverts A. For example, if $A = 0001$, then INV A returns 1000:

A0	B0	S4	S3	S2	S1	S0
A	1	1	0	0	0	0
A	0	1	0	1	0	0

5. [2 points] Draw the logic circuit diagram for the carryout in the full adder:

